

Metabolic Alkalosis

Harold M. Szerlip, Michael Emmett

Metabolic alkalosis is a primary acid-base disorder that increases the serum bicarbonate concentration $[\text{HCO}_3^-]$ above 30 mmol/L,¹ causing the arterial blood $[\text{H}^+]$ to fall and pH to increase into the alkaline range (>7.45). Most laboratories measure the venous $[\text{total CO}_2]$ as a surrogate, which also includes dissolved CO_2 and H_2CO_3 . The $[\text{total CO}_2]$ exceeds the $[\text{HCO}_3^-]$ by about 1 to 2 mmol/L.

Metabolic alkalosis, a very common acid-base disorder, is especially prevalent in critically ill patients.²

RESPIRATORY COMPENSATION

Uncomplicated metabolic alkalosis generates compensatory hypoventilation, which raises Pco_2 , thereby reducing the pH toward the normal range. However, the arterial pH generally remains above 7.45. The magnitude of the hypoventilatory response to metabolic alkalosis is proportional to the degree of $[\text{HCO}_3^-]$ elevation, but the response is quite variable (Fig. 14.1)³⁻⁶ (M.A. Fallahzadeh et al., unpublished observations). Compensation occurs relatively rapidly, within minutes to hours. Mild metabolic alkalosis generates a modest Pco_2 increase (45–50 mm Hg); more severe metabolic alkalosis increases the Pco_2 to the 50 to 55 mm Hg range. Although some patients with severe metabolic alkalosis develop higher Pco_2 values, these patients are usually also very hypokalemic, which can generate muscle weakness and respiratory acidosis.

SERUM CHLORIDE CONCENTRATION AND METABOLIC ALKALOSIS

Abnormal serum chloride concentration $[\text{Cl}^-]$ usually indicates that a water/hydration disorder, an acid/base disorder, or both exist. Water/hydration disorders generate proportional changes of $[\text{Cl}^-]$ and $[\text{Na}^+]$. Therefore, the $[\text{Cl}^-]:[\text{Na}^+]$ ratio remains normal at 1:1.4. $[\text{Cl}^-]$ and $[\text{Na}^+]$ both increase proportionately with dehydration and decrease proportionately with overhydration. This proportional relationship is disrupted when certain acid-base disorders exist (i.e., metabolic alkalosis, hyperchloremic metabolic acidosis, or chronic respiratory disorders). Metabolic alkalosis elevates $[\text{HCO}_3^-]$ and reciprocally reduces $[\text{Cl}^-]$. This disrupts the normal 1:1.4 $[\text{Cl}^-]:[\text{Na}^+]$ ratio. The “anatomic” electrolyte profile, or “Gamblegram” (Fig. 14.2), shows why the increased $[\text{HCO}_3^-]$, which exists with metabolic alkalosis, must be accompanied by an equal reduction of $[\text{Cl}^-]$ (independent of $[\text{Na}^+]$), a reduction of the anion gap, or both.^{7,8} In fact, metabolic alkalosis will commonly slightly increase the anion gap due to increased albumin concentration, increased albumin negative charge density, and accelerated generation of organic acids.^{9,10} To the extent this occurs, the relative $[\text{Cl}^-]$ decrease must be even greater.

An identical electrolyte pattern (increased $[\text{HCO}_3^-]$ and proportionately reduced $[\text{Cl}^-]$) is also generated by compensation for chronic respiratory acidosis. Clinical assessment and blood pH measurement will

indicate the correct diagnosis. Blood pH is increased with metabolic alkalosis and low-normal/reduced with chronic respiratory acidosis. Venous blood pH, although less definitive than arterial pH, can be very helpful; add 0.03 pH units to the venous pH to approximate the arterial pH.¹¹

PATHOGENESIS: GENERATION AND MAINTENANCE OF METABOLIC ALKALOSIS

Seldin and Rector used an extracellular fluid (ECF) volume-centered approach to explain how metabolic alkalosis develops and is maintained.¹² Subsequently, some have proposed that the “ECF volume-centric” approach should be replaced by a “chloride-centric” model,¹³⁻¹⁶ whereas others have challenged this model.^{17,18} From a practical and clinically relevant perspective, ECF volume contraction almost invariably equals “chloride depletion” and vice versa. We will use the traditional ECF volume-centered framework to direct the diagnostic and therapeutic approach to the metabolic alkaloses because it remains very useful and easy to understand.

Normal HCO_3^- Reclamation, Regeneration, and Generation

The metabolism of foods and endogenous substrates produces a large amount of nonvolatile organic acids (i.e., lactic, pyruvic, citric acids) that are oxidized to the volatile acid CO_2 and water. In addition, the typical Western European/American diet generates 80 to 100 mmol/day of nonvolatile strong acids, which cannot be metabolically oxidized (mainly sulfuric, phosphoric, and hydrochloric acids). These strong acids dissociate and release protons that mainly react with HCO_3^- to form H_2CO_3 , which then rapidly dehydrates to CO_2 and H_2O . To the extent this occurs, the serum $[\text{HCO}_3^-]$ falls and is “replaced” by the anions of the generated strong acids, for example, SO_4^{2-} , HPO_4^{2-} , and Cl^- . Acid-base homeostasis must be restored/maintained by the kidney, which filters and/or secretes the anions of these acids, mainly as sodium salts. As the Na salts travel along the tubules, the Na^+ is reabsorbed in exchange for secreted H^+ , which either binds to NH_3 , forming NH_4^+ , or binds to other H^+ binding molecules (i.e., titratable acids), mainly HPO_4^{2-} to generate H_2PO_4^- . Excretion of 80 to 100 mmol of H^+ together with the acid anions (e.g., SO_4^{2-} , HPO_4^{2-} , Cl^-) thereby maintains acid-base equilibrium.

However, before this acid-base restorative process can occur, all the NaHCO_3 filtered by the glomeruli must first be reclaimed (i.e., removed from the filtrate) and returned to the body fluids. About 80% of the normal filtered HCO_3^- load (about 4000–4500 mmol/day) is reclaimed by the proximal tubules via the process of H^+ secretion. The major features of proximal tubule Na^+ reabsorption and H^+ secretion are shown in Fig. 14.3. A large fraction of proximal Na^+ reabsorption occurs via the Na^+-H^+ exchanger 3 (NHE3) in the proximal luminal membrane. This exchange is energized by basolateral membrane

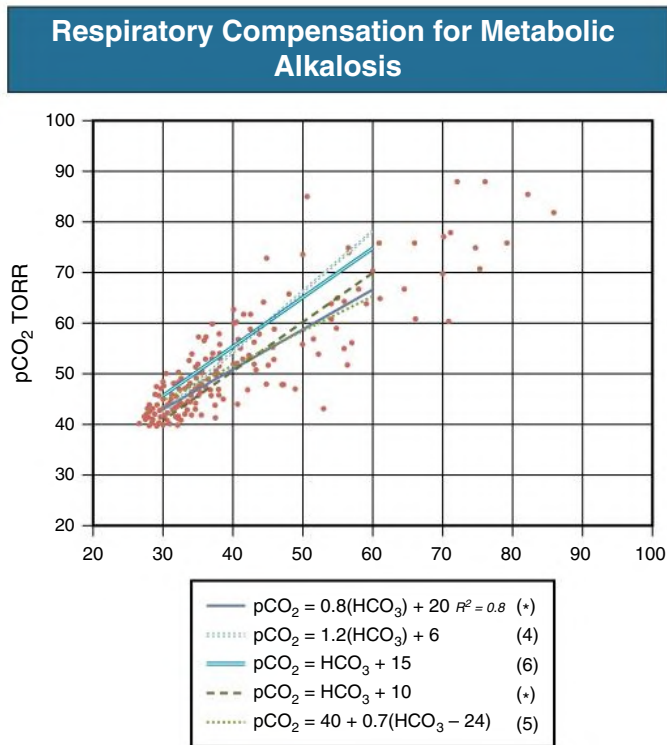


Fig. 14.1 Respiratory Compensation for Metabolic Alkalosis. Simultaneous Pco_2 and $[\text{HCO}_3^-]$ data points derived from a recent comprehensive literature review shows the best-fit linear regression line. Also shown are several commonly used equations that predict the Pco_2 expected for the elevated $[\text{HCO}_3^-]$. (The references are indicated.) We show that the equation $\text{Pco}_2 = [\text{HCO}_3^-] + 10$ (dashed line) is very similar to the best-fit regression line for the HCO_3^- range between 30 and 50 mmol/L. This is a relationship very easy to remember and use. If $[\text{HCO}_3^-]$ exceeds 55 mmol/L, the Pco_2 may increase markedly. This is likely the result of a coexisting respiratory muscle weakness, very likely due to severe hypokalemia that almost always exists. Pco_2 , Partial pressure of carbon dioxide; 1 Torr = 133.32 Pa. Asterisk, Fallahzadeh M.A. et al., unpublished observations. (From Emmet M. Metabolic alkalosis: a brief pathophysiologic review. *Clin J Am Soc Nephrol.* 2020;15[12]:1848–1856.)

Na^+ , K^+ -ATPase, which reduces intracellular Na^+ and generates a negative intracellular electrical charge. The resulting strong electrochemical gradient drives Na^+ from the lumen into the cell in exchange for H^+ entering the lumen. Additionally, ATP-energized H^+ pumps in the luminal membrane of the proximal tubule are responsible for about 30% of proximal H^+ secretion.

Each H^+ that enters the proximal tubule lumen leaves 1 HCO_3^- in the cytoplasm, which moves across the basolateral membrane to enter the ECF. The HCO_3^- movement from within the cell to the ECF occurs largely via the NBCe1A transporter (which moves 1 Na^+ and 3 HCO_3^- ions). The net effect of the secretion of 1 H^+ molecule is the disappearance of 1 HCO_3^- molecule from the luminal fluid and appearance of 1 HCO_3^- molecule in the ECF.

About 20% of the filtered HCO_3^- load escapes proximal tubule reclamation and is delivered downstream, where most is reabsorbed by the thick ascending limb of Henle. Proton secretion here is largely accomplished by the same Na^+ - H^+ exchanger (NHE3) as exists in the proximal tubule. The small fraction of HCO_3^- delivered more distally is reabsorbed by mechanisms shown in Fig. 14.4.

After HCO_3^- has largely disappeared from the lumen in distal tubules/collecting ducts, further H^+ secretion generates “new” HCO_3^- that restores, or “regenerates,” the HCO_3^- that was decomposed by

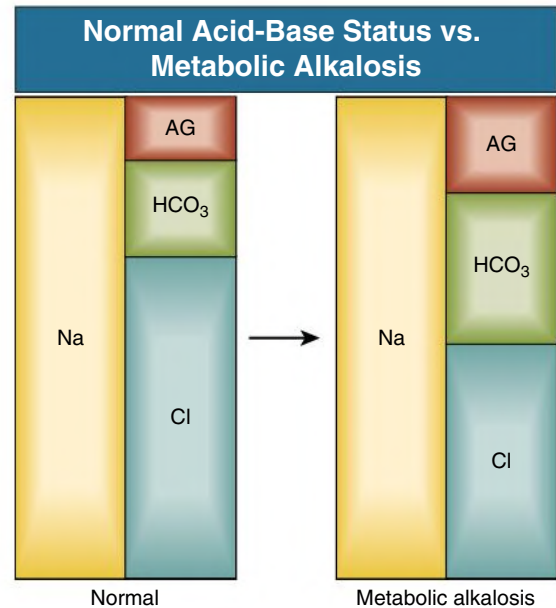


Fig. 14.2 Normal Acid-Base Status and Metabolic Alkalosis. The three major serum electrolytes are shown in the form of Gamblegrams. Note that when $[\text{HCO}_3^-]$ increases and the anion gap (AG) also increases slightly (which occurs with most forms of metabolic alkalosis), $[\text{Cl}^-]$ must fall, and the $[\text{Cl}^-]:[\text{Na}^+]$ ratio will fall below its normal 1:1.4 relationship.

sulfuric, phosphoric, and hydrochloric acids, for example. Also, the quantity of HCO_3^- that has been lost in the stool must also be regenerated. If 80 to 100 mmol/day of nonvolatile acids have been generated, then 80 to 100 mmol/day of H^+ (plus any HCO_3^- lost in stool) must be excreted by the kidneys to maintain acid-base equilibrium.

As HCO_3^- disappears from the tubular fluid, the pH falls. However, the minimum tubule fluid/urine pH cannot be reduced below about 4.5, which represents only 0.03 mmol/L of free H^+ (and therefore only 0.03 mmol/L of generated HCO_3^-). Therefore, virtually the entire H^+ load excreted in the urine must be bound to buffers in the urine: HPO_4^{2-} , which is titrated to H_2PO_4^- , represents most of the titratable acid in urine, and NH_3 binds to H^+ to form NH_4^+ .

Metabolic Alkalosis: Source of Excess HCO_3^-

Metabolic alkalosis usually indicates an accumulation of “excess” HCO_3^- in the body. The HCO_3^- source can be exogenous, endogenous, or both.

Exogenous Sources

Exogenous sources include Na^+ or K^+ HCO_3^- or salts of HCO_3^- precursors (organic anions such as lactate, acetate or citrate, which generate HCO_3^- when completely oxidized). These salts may be ingested/absorbed and/or infused. (Box 14.1).

Endogenous Sources

The two most important potential sources of endogenous of HCO_3^- are the stomach and kidney, as described below.

Stomach

HCO_3^- is added to the ECF whenever HCl is secreted into gastric lumen. Essentially, 1 HCO_3^- ion replaces 1 Cl^- in the ECF. However, net HCO_3^- accumulation in the ECF requires the secreted HCl to be lost from the body, usually via vomiting and/or gastric suction (for further explanation, see the section Classic Example: Gastric Alkalosis).

Major Proximal Tubule Cellular and Luminal Events in HCO_3^- Reclamation

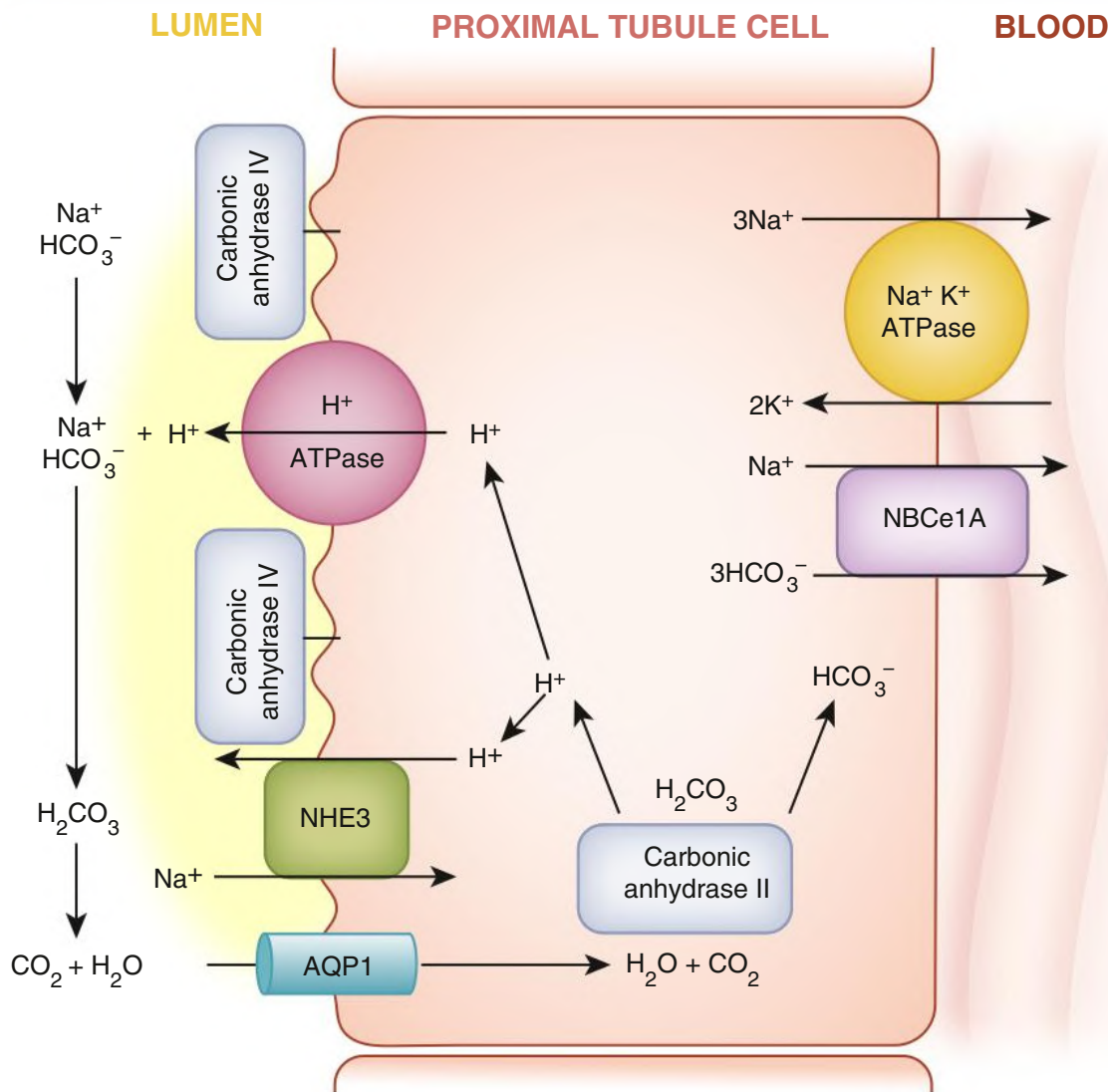


Fig. 14.3 Major Proximal Tubule Cellular and Luminal Events That Participate in HCO_3^- Reclamation. Intracellular H_2O combines with CO_2 to generate H_2CO_3 , which rapidly dissociates to H^+ and HCO_3^- ions. These reactions are catalyzed by the cytoplasmic enzyme carbonic anhydrase II. Na^+/K^+ -ATPase in the basolateral membrane creates a steep Na^+ electrochemical gradient that energizes Na^+ reabsorption via the Na^+/H^+ exchanger 3 (NHE3). When H^+ moves into the lumen, it reacts with filtered HCO_3^- to form H_2CO_3 . The H_2CO_3 dehydrates to H_2O and CO_2 , a reaction catalyzed by intraluminal carbonic anhydrase IV (this enzyme is anchored by a glycosylphosphatidylinositol tail to the luminal membrane). The generated CO_2 flows into the cell, largely via the aquaporin 1 (AQP1) channel. The HCO_3^- ions generated within the cells move across the basolateral membrane into peritubular capillaries, primarily via an $\text{Na}^+/\text{HCO}_3^-$ cotransporter (the electrogenic NBCe1A transporter is a product of the Solute Carrier Family 4 Member A4 [*SLC4A4*] gene). The net effect of the secretion of 1 H^+ molecule and reabsorption of 1 Na^+ molecule is the addition of 1 NaHCO_3 molecule to the extracellular fluid and the disappearance of 1 NaHCO_3 molecule from the lumen. In addition, a smaller component of proximal tubule H^+ secretion (HCO_3^- reclamation) is accomplished via a V-type H^+ -ATPase pump complex. Under normal conditions, about 80% of the filtered NaHCO_3 is reclaimed in the proximal tubule.

Kidney

Normal excretion of H^+ in the urine (as NH_4^+ and titratable acid) generates the quantity of HCO_3^- required to replace the HCO_3^- decomposed by nonvolatile acids derived from dietary metabolism plus any HCO_3^- lost in the stool. To the extent the kidney generates additional HCO_3^- above these replacement requirements, that quantity represents

“excess” HCO_3^- and can generate metabolic alkalosis. “Excess” kidney HCO_3^- generation generally occurs when the following conditions coexist:

1. The more distal kidney tubules/ducts are stimulated to avidly reabsorb Na^+ (e.g., because aldosterone activity is high), and

Principal Cells and Type A and Type B Intercalated Cells

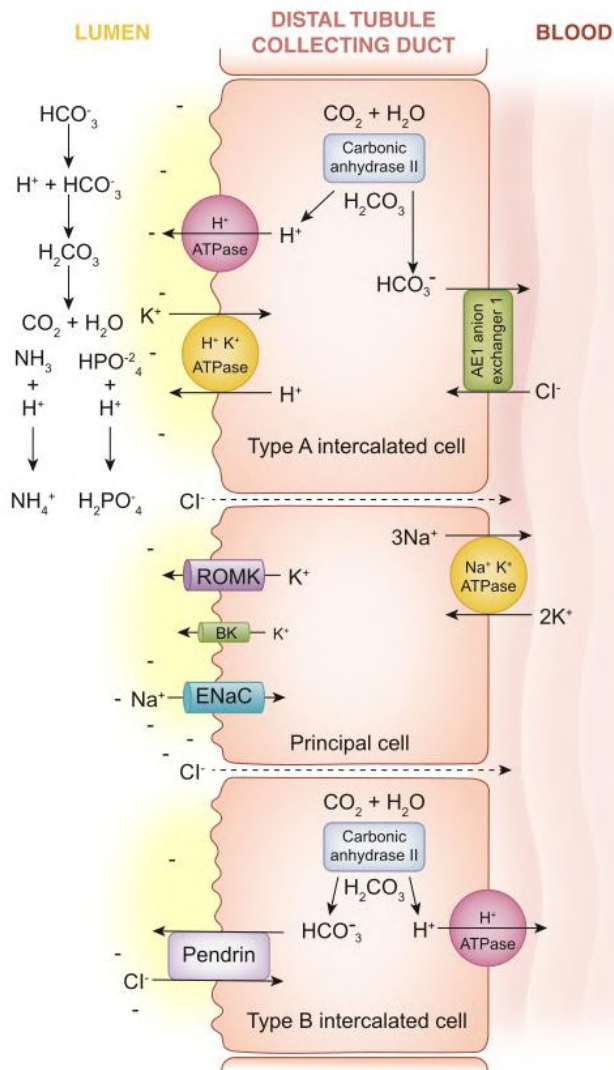


Fig. 14.4 Principal Cells and Type A and Type B Intercalated Cells. These cells are located in the late distal convoluted tubule, the connecting tubule, and cortical collecting duct. Principal cell transport is energized primarily by electrogenic Na^+ , K^+ -ATPase pumps in the basolateral membrane. The resulting low intracellular $[Na^+]$ and negative intracellular potential difference combine to create a large gradient for Na^+ to move from the lumen into these cells. Flow occurs mainly through the epithelial sodium channel (ENaC, which is composed of α , β , and γ subunits, each a product of a different gene). Inward flow of Na^+ ions is more rapid than the sum of outward movement of K^+ (mainly through the renal outer medullary potassium [ROMK] channel) and inward flow of Cl^- (mainly via the paracellular space). Therefore, the Na^+ influx generates a negative potential difference (PD, approximately -40 mv) in the lumen. The H^+ secreted by the type A intercalated cells into the lumen initially reacts with any HCO_3^- remaining in the lumen; this represents HCO_3^- reclamation. As the luminal fluid pH falls, additional secreted H^+ combines with buffers such as HPO_4^{2-} and NH_3 to generate urine titratable acid (i.e., $H_2PO_4^{2-}$) and NH_4^+ , which creates equimolar quantities of systemic HCO_3^- . See Fig. 14.5 for a more detailed description of intercalated cell ion transport.

BOX 14.1 Metabolic Alkalosis: Bicarbonate Sources and Mechanisms of Generation

- Ingest/absorb and/or infuse $NaHCO_3$ and/or $NaHCO_3$ precursors, such as Na acetate, Na citrate, or Na gluconate.
- Generation of excess HCO_3^- by distal renal tubules/collecting ducts (increased H^+ secretion).
- Generous delivery of NaCl (or other Na salts, such as Na_2SO_4 or Na penicillin) to distal tubules/collecting ducts, which are actively reabsorbing Na^+ (and therefore secreting K^+ and H^+).
- K^+ depletion, shifting H^+ into cells, thereby generating extracellular fluid HCO_3^- .
- Remove HCl from body (vomiting/nasogastric suction/chloride-rich diarrhea).

2. Salt and volume delivery to these distal reabsorbing sites is relatively large.

Several examples of kidney bicarbonate overproduction include:

1. Primary hyperaldosteronism and other disorders, which mimic primary hyperaldosteronism (especially when a high-salt diet is ingested) (Box 14.2)
2. Loop and/or thiazide diuretics
3. Certain inherited kidney salt-wasting syndromes (i.e., Bartter, Gitelman, and related syndromes)
4. The infusion of Na^+ salts of poorly absorbed anions (e.g., HPO_4^{2-} , SO_4^{2-} , penicillin) when distal tubule Na^+ reabsorption is stimulated by mineralocorticoids and/or volume contraction^{19,20}

Another important endogenous source of “excess” ECF HCO_3^- is the movement of K^+ out of cells into the ECF (in response to hypokalemia), which is partially balanced by movement of H^+ into cells. This generates HCO_3^- in the ECF.^{21,22} The ECF $[HCO_3^-]$ can also increase when ECF volume contracts around a fixed quantity of HCO_3^- (see the section Classic Example: Gastric Alkalosis).²³

Maintenance of Metabolic Alkalosis

If metabolic alkalosis develops and glomerular filtration rate (GFR) is not markedly reduced, correction of the alkalosis should be relatively straightforward: merely excrete a large fraction of the filtered HCO_3^- (which will be supernormal because of the high serum/filtered $[HCO_3^-]$). A brisk HCO_3^- diuresis would then reduce the $[HCO_3^-]$ and restore normal acid-base status. This obviously has not occurred when metabolic alkalosis persists. Why has a brisk HCO_3^- diuresis not developed to rapidly restore normal acid-base status? The answer to this question depends on the underlying cause of the alkalosis, and the precise explanations continue to be refined and debated. The major mechanisms responsible for the maintenance of metabolic alkalosis are shown in Box 14.3.

Normal individuals can ingest, and their kidneys will efficiently excrete, up to 1000 mmol/day of $NaHCO_3$ over several weeks with only a minimal increase in serum $[HCO_3^-]$.²⁴ Consequently, when metabolic alkalosis develops and persists despite a relatively normal GFR, this indicates the kidney is avidly reclaiming HCO_3^- at a supernormal rate. Most patients with metabolic alkalosis have both increased proximal and distal HCO_3^- reabsorption. The major factors responsible increasing proximal HCO_3^- are reduced intravascular, or effective intraarterial, blood volume, and hypokalemia.¹² (Metabolic acidosis and chronic respiratory acidosis also stimulate proximal tubule HCO_3^- reabsorption, but these disorders are not relevant to the current discussion.)

Metabolic alkalosis is also usually associated with accelerated distal HCO_3^- reabsorption and generation. Generous distal Na^+ delivery, combined with avid distal Na^+ reabsorption (e.g., because of high aldosterone levels), accelerates distal H^+ and K^+ secretion. This occurs

BOX 14.2 Causes of Metabolic Alkalosis**ECF Volume Contracted: Urine $[\text{Cl}^-] < 20 \text{ mmol/L}$**

- Gastric alkalosis: vomiting/nasogastric suction
- Chloride-rich diarrhea (congenital chloridorrhea)
- Status post–chronic hypercapnia (acute reversal of chronic respiratory acidosis)
- Cystic fibrosis with major sweating
- Some villous adenomas
- Thiazide or loop diuretics after their renal tubule diuretic effect has dissipated

ECF Volume Expanded: Urine $[\text{Cl}^-] > 20 \text{ mmol/L}$

- Primary hyperaldosteronism (unilateral adenoma/bilateral hyperplasia/glucocorticoid-sensitive hyperaldosteronism)
- Severe Cushing syndrome (especially with ectopic adrenocorticotrophic hormone secretion)
- Exogenous mineralocorticoids
- Reduced 11- β (OH)-steroid dehydrogenase activity
 - Chronic licorice/carbenoxolone ingestion
 - Congenital apparent mineralocorticoid excess syndrome (inactivating mutation of 11- β hydroxysteroid dehydrogenase (type 2))
- Renin-secreting tumors
- Some forms of congenital adrenal hyperplasia
 - 11- β hydroxylase deficiency
 - 17- α hydroxylase deficiency
- Liddle syndrome (persistent open epithelial sodium channel–renal tubule epithelial channels)

ECF Volume Contracted: But Urine $[\text{Cl}^-] > 20 \text{ mmol/L}$ (Generally Indicates a Renal Tubule NaCl Reabsorptive Defect)

- Thiazide and/or loop diuretics actively inhibiting renal tubule NaCl reabsorption
- Bartter syndromes (defective Na reabsorption in loop of Henle, a furosemide-like lesion)
- Gitelman syndrome (defective Na reabsorption by the neutral NaCl transporter, a thiazide-like lesion)

Metabolic Alkalosis: Other

- Very severe potassium deficiency
- Milk (calcium) alkali syndrome
- NaHCO_3 loads in patients with markedly reduced glomerular filtration rate.
- Refeeding after prolonged fasting

ECF, Extracellular fluid.

in large part because distal tubule Na^+ reabsorption, mainly via the epithelial sodium channel (ENaC) in principal cells, generates a lumen negative electric potential (potential difference $[\text{PD}] = -40 \text{ mV}$), which drives paracellular and transcellular anion (mostly Cl^-) reabsorption and enhances the secretion of H^+ and K^+ . Also, multiple neurohormonal stimuli that increase distal Na^+ absorption also increase type A intercalated cell activity²⁵ (Fig. 14.4).

Three different types of intercalated cells are shown in Fig. 14.5. The type B intercalated cell secretes HCO_3^- in exchange for Cl^- and therefore contributes to the correction of metabolic alkalosis. However, generous distal delivery of Cl^- is required to enable this cell to secrete major quantities of HCO_3^- , and Cl^- delivery is usually reduced when ECF, or effective intraarterial volume, is reduced. Intercalated cells also contribute to NaCl reabsorption and volume regulation as discussed in Fig. 14.5.^{25–31}

Of course, if GFR is markedly reduced, the filtered HCO_3^- load will fall sharply. However, metabolic alkalosis does not usually develop

BOX 14.3 Metabolic Alkalosis: Mechanisms of Maintenance

- Increased proximal renal tubule HCO_3^- reclamation
 - True extracellular fluid contraction or “effective intraarterial” volume contraction
 - K^+ depletion
- Continuous or intermittent generation of new HCO_3^-
 - In the distal kidney tubules and collecting ducts
 - Gastric HCl losses
- Exogenous alkali
- Reduced NaHCO_3 filtration due to kidney failure

in patients with very low GFR. These patients more often develop metabolic acidosis. Nonetheless, occasionally metabolic alkalosis does develop, and under these conditions the reduced GFR impairs excretion of the HCO_3^- load. A reduced GFR is a major contributor to the maintenance of metabolic alkalosis in patients with milk-alkali or calcium-alkali syndrome^{32,33} and when patients with severe kidney dysfunction ingest bicarbonate salts, bicarbonate generating salts, vomit, or require nasogastric (NG) suction.³⁴

METABOLIC ALKALOSIS: ETIOLOGIC CLASSIFICATION BASED ON ECF VOLUME STATUS

The generation, maintenance, and resolution of several prototypical types of metabolic alkalosis in each ECF volume status category are described.

ECF Volume Contracted With Persistently Reduced Urine $[\text{Cl}^-]$ **Classic Example: Gastric Alkalosis**

Generation. Gastric fluid osmolality is about 300 mOsm/L with a $[\text{Cl}^-]$ of about 150 mmol/L and a similar total cation concentration of about 150 mmol/L. The $[\text{H}^+]$ varies between 40 and 140 mmol/L, $[\text{K}^+]$ between 10 and 15 mmol/L, and $[\text{Na}^+]$ makes up the balance.³⁵ Secretion of HCl (via gastric type H^+/K^+ ATPase and Cl^- channels) into the gastric lumen adds an equal quantity of HCO_3^- to the ECF. (The net effect of HCl secretion is the “replacement” of ECF Cl^- by HCO_3^- .) Normally, gastric HCl secretion does not result in metabolic alkalosis because the secreted HCl is not externally lost from the body. The HCl secreted into the gastric lumen usually enters the small bowel where the H^+ reacts with HCO_3^- , mainly secreted by the pancreas (with a small component derived from bile and intestinal epithelium). This reaction generates CO_2 and water in the small bowel lumen. Because the quantity of HCO_3^- that enters the small bowel lumen and reacts with H^+ equals the quantity of H^+ delivered from the stomach, these two processes neutralize one another so that neither alkalosis nor acidosis develops. However, if the HCl secreted into the stomach is removed from the body, via vomiting or aspiration, this prevents the HCl from reaching the small bowel. Consequently, HCO_3^- is either not secreted into the intestinal lumen, for neutralization by HCl, or is reabsorbed. The net effect is the addition of a gastric derived load of HCO_3^- to the ECF and loss of an equal quantity of Cl^- from the body. This represents the initial source of HCO_3^- in patients with “gastric alkalosis.”

Later in the development of gastric alkalosis, hypokalemia (generated by renal K^+ loss) results in systemic K^+/H^+ cell shifts, which generate additional ECF HCO_3^- (described later).

Intercalated Cells

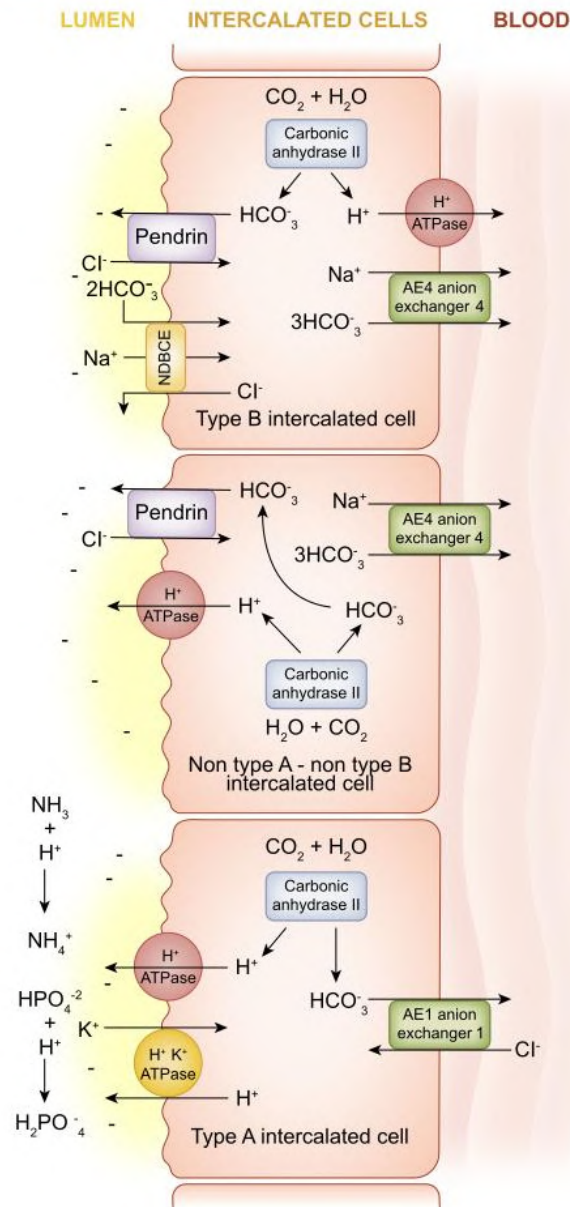


Fig. 14.5 Intercalated Cells: Three different intercalated cell types have been identified: type A intercalated cells, type B intercalated cells, and non-type A/non-type B intercalated cells. Each is rich in carbonic anhydrase II and is capable of generating abundant HCO_3^- and H^+ from H_2CO_3 . **Type A intercalated cells:** H^+ is secreted into the lumen mainly via V-type H^+ -ATPase and to a smaller extent via H^+ -K $^+$ -ATPase. The generated cytoplasmic HCO_3^- moves into the peritubular capillary in exchange for Cl^- , via anion exchanger 1 (AE1; a product of the Solute Carrier Family 4 Member 1 [SLC4A1] gene). This transporter is also called the *erythrocyte membrane protein band 3* when it is located in red blood cells. **Type B intercalated cells:** H^+ is secreted into the peritubular capillary by V-type H^+ -ATPase in the basolateral membrane (the same proton transporter as found in type A intercalated cells but in the opposite, or luminal, membrane). The HCO_3^- generated in this cell is secreted into the lumen via an anion exchanger named pendrin (a product of the Solute Carrier Family 26 Member A4 [SLC26A4] gene), which exchanges 1 HCO_3^- for 1 Cl^- . Pendrin is a distinct and different exchanger than the AE1 present in the basolateral membrane of the type A cells. A $\text{Na}^+/3\text{HCO}_3^-$ cotransporter AE4 (anion exchanger 4; a product of the Solute Carrier Family 4 Member A4 or [SLC4A4] gene) is present in the basolateral membrane. These cells also have the sodium-driven bicarbonate chloride exchanger (NDBCE; a product of the Solute Carrier Family 4 Member A8 [SLC4A8] gene) in the luminal membrane (see later discussion). **Non-type A/non-type B intercalated cells:** this is the third type of intercalated cell. Both a V-type H^+ -ATPase, pumping H^+ , and pendrin, exchanging HCO_3^- for Cl^- coexist in the lumen membrane. AE4 is a major $\text{Na}^+/3\text{HCO}_3^-$ transporter in the basolateral membrane, moving these ions into the peritubular capillary. Although intercalated cells were originally identified as major kidney acid-base-regulating cells, it is now clear that these cells also play an important role in salt and volume regulation.^{32,33} The NDBCE in the luminal membrane of the type B intercalated cells is an electrically neutral ion exchanger moving 1 Na^+ and 2 HCO_3^- ions into the cell while moving 1 Cl^- into the lumen. The net effect of two pendrin cycles (2 HCO_3^- ions enter the lumen in exchange for uptake of 2 Cl^- ions) and 1 NDBCE cycle (2 HCO_3^- ions and 1 Na^+ ion enter the cell and 1 Cl^- ion enters the lumen) has the net effect of the reabsorption of 1 NaCl molecule. These acid-base and salt reabsorption interactions of intercalated cells may explain why some patients with Pendred syndrome, a genetic defect of pendrin, develop severe salt depletion and metabolic alkalosis when treated with thiazide diuretics.³⁴ (Pendrin is also a major transporter in the ear, and patients with Pendred syndrome are usually congenitally deaf.)

Maintenance. Initially, as HCO_3^- is added to the ECF and slightly raises the $[\text{HCO}_3^-]$, as a result of vomiting or gastric suction, most is filtered and excreted by the kidneys largely as NaHCO_3 . The loss of gastric fluid combines with kidney loss of NaHCO_3 and fluid to produce ECF volume contraction, and this activates a neurohormonal cascade that stimulates the kidneys to avidly retain salt and fluid. Secondary hyperaldosteronism develops and results in the conversion of much of the NaHCO_3 delivered to the distal tubules to KHCO_3 .³⁶

Systemic hypokalemia develops causing K^+ to shift out of cells in exchange for ECF H^+ entering cells. This generates ECF HCO_3^- .^{21,22} This K^+/H^+ shift acidifies kidney tubule cells, which further stimulates H^+ secretion and thereby drives kidney HCO_3^- reclamation and generation. Systemic hypokalemia also reduces pendrin activity and type B intercalated cell density, which reduces HCO_3^- secretion and contributes to the maintenance of metabolic alkalosis.^{30,31}

Extracellular fluid volume contraction stimulates both proximal and distal Na^+ reabsorption. In the proximal tubules, this increases HCO_3^- reclamation. In the distal tubules, Na^+ reabsorption increases both H^+ and K^+ secretion. K^+ depletion further increases H^+ secretion by type A intercalated cells via both H^+ ATPase and H/K ATPase pumps (Figs. 14.3 and 14.4) and increases kidney NH_3 generation and excretion.^{37,38} These effects all combine to increase kidney HCO_3^- reclamation and generation.

During the maintenance phase of gastric metabolic alkalosis, the urine electrolyte pattern typically fluctuates. Extracellular fluid volume contraction stimulates avid reabsorption of filtered Na^+ , HCO_3^- , Cl^- , and water. This usually generates concentrated urine (an antidiuretic hormone effect) with low Na , Cl , and K concentrations and a relatively acid pH (denoted as *paradoxical aciduria* because systemic metabolic alkalosis exists). However, intermittently (e.g., immediately after loss of a bolus of HCl -rich gastric fluid), serum $[\text{HCO}_3^-]$ acutely increases, and for a limited period of time, the larger load of filtered HCO_3^- cannot be completely reclaimed despite the multiple neurohormonal factors generated by volume contraction. When this occurs, the urine excretion of $[\text{HCO}_3^-]$, $[\text{Na}^+]$, $[\text{K}^+]$, and pH all temporarily increase. This reduces the serum $[\text{HCO}_3^-]$ and further contracts ECF volume, and after a brief period of time the filtered load of Na^+ and HCO_3^- can again be completely reabsorbed. The urine electrolyte concentrations and urine pH again fall. However, it is very important to note that throughout these cyclic variations in urine $[\text{Na}^+]$, $[\text{K}^+]$, $[\text{HCO}_3^-]$, and pH, urine $[\text{Cl}^-]$ remains persistently low (because the intermittent loss of HCO_3^- in the urine mandates loss of Na^+ and/or K^+ but not Cl^-). Thus, a low urine $[\text{Cl}^-]$ (<20 mmol/L) generally indicates the ongoing response to reduced ECF/intraarterial blood volume (or very low Cl^- intake). Development of these urine electrolyte profiles presume that kidney tubule function is relatively intact and the absence of exogenous diuretic activity.

Resolution. The factors responsible for kidney HCO_3^- retention in these forms of metabolic alkalosis are largely reversed by adequate ECF volume expansion (NaCl infusion) and K^+ repletion. Restoration of ECF volume is associated with a rising urine $[\text{Cl}^-]$ and the development of a NaHCO_3 diuresis. Increased distal Cl^- delivery also stimulates HCO_3^- secretion in exchange for Cl^- via the pendrin exchanger in Type B intercalated cells. Systemic K^+ replacement is required to restore K^+ homeostasis. Some of the administered K^+ moves into cells in exchange for H^+ moving into the ECF. This simultaneously reduces the ECF $[\text{HCO}_3^-]$ and increases intracellular pH.

The term *contraction alkalosis* has been used to describe several different conditions in which the ECF “contracts” around a relatively fixed quantity of HCO_3^- . Although gastric alkalosis is an ECF volume

contracted condition and the ECF contraction contributes importantly to both its generation and maintenance, the major origin of the increased blood $[\text{HCO}_3^-]$ is not contraction of the ECF per se but rather generation of HCO_3^- owing to gastric HCl loss and cellular H^+/K^+ shifts.^{21-23,36}

Conversely, although ECF expansion with NaCl -rich fluids does directly dilute the ECF $[\text{HCO}_3^-]$ to a small degree, the corrective action of NaCl in these patients is mainly the result of ECF volume expansion and resultant kidney HCO_3^- excretion.

Reducing or stopping the loss of gastric HCl is of course required to reverse the process at its initiation point. If the exogenous loss of gastric fluid losses cannot be stopped, then reducing the gastric fluid $[\text{HCl}]$ concentration with an H^+ blocker or proton pump inhibitor can be helpful.³⁹

Box 14.2 lists the most common forms of metabolic alkalosis associated with ECF volume contraction. The urine $[\text{Cl}^-]$ is typically reduced to less than 20 mmol/L in each of these disorders.

ECF Volume Expanded

Classic Example: Primary Mineralocorticoid Excess Syndromes

Primary hyperaldosteronism is a condition of autonomous, or inappropriately upregulated, aldosterone secretion. This usually generates ECF volume expansion, hypertension, hypokalemia, and metabolic alkalosis. A unilateral adrenal adenoma secreting aldosterone is the prototypical cause of this disorder, but many conditions can duplicate or mimic the clinical and electrolyte and acid-base pathophysiology of a primary aldosterone-secreting adenoma (Box 14.2) (see Chapter 39).

ECF Volume Regulation of Renin and Aldosterone (Normal Physiology)

Extracellular fluid volume contraction in normal individuals reduces the GFR and sharply increases the reabsorption of NaCl and NaHCO_3 in the proximal tubules. Volume contraction also generates secondary hyperaldosteronism (high aldosterone activity driven by high renin and angiotensin II activity). Reabsorption of Na^+ in late distal tubules/collecting ducts is largely accomplished by the principal cells via ENaC channels in the luminal membranes (Fig. 14.4). When these channels are present in the luminal membrane and in an open state, the electrochemical gradient favors reabsorption of Na^+ , and this cation flux generates a lumen negative electrical potential (about -40 mV) that drives both Cl^- reabsorption and also secretion of K^+ and H^+ . Normally, ECF volume contraction generates a coordinated series of proximal and distal actions so that the potent aldosterone-driven stimulus to reabsorb Na^+ in the distal tubules and collecting ducts is linked to a fall in GFR and increased proximal tubule salt reabsorption. These proximal tubule events sharply reduce distal salt and water delivery. The net effect is that reduced Na^+ and volume delivery to the distal aldosterone-sensitive tubule sites acts to blunt the magnitude of distal Na^+ reabsorption and the indirectly linked secretion of K^+ and H^+ . Thus, metabolic alkalosis and hypokalemia do not usually develop despite physiologic secondary hyperaldosteronism.

An opposite series of events develop in response to ECF volume expansion in normal individuals. The GFR increases, proximal salt and water reabsorption fall, and distal delivery of salt and water is generous. Volume expansion reduces renin, angiotensin II, and aldosterone levels. Now generous distal salt and water delivery exists while aldosterone activity is minimal. Distal Na^+ reabsorption, and indirectly linked K^+ and H^+ secretion, is moderate. Thus, metabolic

acidosis and hyperkalemia do not develop despite physiologic secondary hypoadosteronism.

These coordinated interactions describe the normal reciprocal physiologic balance that exists between the magnitude of proximal tubule salt and water reabsorption, distal delivery of salt and water, and the neurohormonal stimulation (largely via aldosterone) of distal Na^+ reabsorption and indirectly linked K^+ and H^+ secretion. This exquisite normal reciprocal balance is disrupted by autonomous hypersecretion of aldosterone.^{12,40}

Generation. Autonomous hyperaldosteronism generates an inappropriate degree of distal Na^+ reabsorption, which expands the ECF, raises the GFR, and reduces proximal tubule salt and water reabsorption. This results in generous distal salt and water delivery. This occurs in the setting of persistently high aldosterone activity. These conditions result in the unphysiologic combination of high aldosterone activity linked with high distal tubule salt and water delivery. This pathophysiologic combination generates a high rate of distal Na^+ reabsorption and indirectly linked generous K^+ and H^+ secretion. Excess excretion of K^+ and H^+ exceeds physiologic requirements and produces hypokalemia and metabolic alkalosis.

The development of hypokalemia and K^+ depletion contribute importantly to additional systemic and kidney HCO_3^- generation. K^+ shifts out of cells in exchange for H^+ generating ECF HCO_3^- .²² Systemic K^+ depletion also increases kidney H^+ secretion and ammonia generation and excretion (in part by reducing the pH of kidney tubule cells).^{25,27} All these effects enhance kidney HCO_3^- reclamation and generation.

Maintenance phase. Expansion of the ECF usually increases the GFR, reduces proximal tubule salt and volume reabsorption, and decreases HCO_3^- reclamation. ECF expansion normally also reduces renin, angiotensin II, and aldosterone levels. Consequently, with “physiologic” ECF volume expansion, distal Na^+ reabsorption and K^+ and H^+ secretion remain modest despite high delivery rates. However, autonomous aldosterone secretion combines generous distal salt delivery with inappropriately high levels of distal Na^+ reabsorption and indirectly linked K^+ and H^+ excretion. Hypokalemia and K^+ depletion develop and contribute importantly to both ongoing kidney HCO_3^- reclamation and both kidney and systemic HCO_3^- generation and via K^+/H^+ cell shifts and cellular acidification. Hypokalemia also increases H^+/K^+ -ATPase activity in type A intercalated cells (Figs. 14.4 and 14.5).^{25,27} Additionally, aldosterone increases distal salt reabsorption via a sequence of pendrin-related events as described in Fig. 14.5.²⁸

During the maintenance phase of various states of autonomous hyperaldosteronism, and related pathophysiologic conditions, urine electrolytes reflect the individual’s salt intake. Thus, the urine $[\text{Cl}^-]$ will generally be greater than 20 mmol/L, unless the patient is ingesting a very low NaCl diet.

Recovery phase. Successful resection of an adrenal aldosterone-secreting adenoma should theoretically reverse the entire syndrome. In lieu of surgery, drugs that block the action of aldosterone can be very helpful. The physical and biochemical manifestations of primary hyperaldosteronism can also be markedly ameliorated by ingestion of a low-salt diet. This reduces distal salt delivery and thereby blunts H^+ and K^+ secretion and excretion. Conversely, the physical findings and electrolyte abnormalities are exacerbated by a high-salt diet.^{40,41} Analogously, other mineralocorticoid excess syndromes and mineralocorticoid excess-like syndromes are improved or “cured” by eliminating the source of the problem and/or treated by blocking the downstream pathophysiology. However, when hypertension has existed for a long period of time, this may persist as a result of the development of structural vascular pathology.

ECF Volume Contracted With Intermittently Reduced Urine $[\text{Cl}^-]$

Classic Example: Thiazide and/or Loop Diuretics

Thiazide and/or loop diuretics often generate hypokalemia and metabolic alkalosis. Despite a relatively contracted ECF and effective arterial blood volume, the generation and maintenance mechanisms of these conditions have many similarities to the ECF volume-expanded condition of primary hyperaldosteronism.⁴⁰ That is because distal tubule salt and volume delivery are high when diuretics are inhibiting reabsorption. Therefore, they generate the pathophysiologic combination of high distal salt/volume delivery and high levels of aldosterone activity (in this case due to secondary hyperaldosteronism).

Generation. Inhibition of the neutral $\text{Na}/\text{K}/2\text{Cl}$ cotransporter in the thick limb of Henle by loop diuretics and/or inhibition of the neutral Na/Cl cotransporter in the diluting segment by thiazide diuretics increases NaCl and volume delivery to more distal sites. Diuretics also increase renin, angiotensin II, and aldosterone levels, generating secondary hyperaldosteronism. In the absence of diuretics, secondary hyperaldosteronism is typically associated with reduced distal salt and volume delivery, and this limits the magnitude of distal Na^+ reabsorption (and thereby H^+ and K^+ secretion). However, diuretic activity combines secondary hyperaldosteronism with generous distal tubule NaCl and volume delivery. Therefore, enhanced distal Na^+ reabsorption via principal cell ENaCs is linked with generous distal Na^+ and volume delivery, accelerating distal H^+ and K^+ secretion. This pathophysiologic combination generates metabolic alkalosis and hypokalemia. Hypokalemia also adds additional ECF HCO_3^- via cellular H^+/K^+ exchange described above.

During periods of diuretic activity, urine $[\text{Na}^+]$ and $[\text{Cl}^-]$ are both high. However, diuretics are typically administered intermittently, so that periods of diuretic activity cycle with periods of inactivity and recovery. During the “off-diuretic” phases, avid proximal kidney salt and water reabsorption markedly reduces distal NaCl delivery. This minimizes principal cell Na^+ reabsorption and distal K^+ and H^+ secretion. During these off-diuretic periods, urine $[\text{Cl}^-]$ and $[\text{Na}^+]$ fall to low levels, reflecting the patient’s relatively contracted ECF volume status. Then, when the diuretic is again ingested or administered, the urine $[\text{Cl}^-]$ and $[\text{Na}^+]$ increase. Thus, the concentration of these urine electrolytes cycles up and down depending on the presence or absence of diuretic activity.

Recovery phase. When diuretics are discontinued, distal delivery of salt and volume fall, and this reduces kidney tubule HCO_3^- generation. However, metabolic alkalosis will not resolve until the potent stimuli, which are accelerating proximal and distal salt reabsorption, are eliminated. Therefore, if diuretics were initiated to treat avid salt retention (e.g., due to heart failure, hepatic cirrhosis), the metabolic alkalosis will generally persist until the underlying pathology can be ameliorated. Reversing hypokalemia and K^+ depletion is also critical for the amelioration or reversal of metabolic alkalosis. If the clinical situation mandates continuing diuretics despite metabolic alkalosis, the addition or substitution of diuretics, which blunt distal tubule/collecting duct Na^+ reabsorption (i.e., potassium sparing diuretics such as triamterene or spironolactone) and/or “acidifying” diuretics, which block proximal tubule HCO_3^- reclamation (acetazolamide), can be very helpful. However, acetazolamide will generate marked K^+ wasting, so aggressive K^+ supplementation is often required when this diuretic is initiated.⁴²

All three types of intercalated cells located in the distal tubule/collecting ducts play a major role in both acid/base regulation and also

volume regulation and salt balance.^{43,44} These cells may be especially important in moderating the development of metabolic alkalosis in patients receiving thiazide diuretics⁴⁵ (Fig. 14.5).

OTHER METABOLIC ALKALOSSES

Bartter and Gitelman syndromes and their related disorders are genetic diseases that result in impaired kidney tubule salt reabsorption. Therefore, these diseases have many similarities to persistent diuretic action. However, unlike patients who receive diuretics, these patients never experience an “off-drug” period. Therefore, these disorders are characterized by persistent, high urine $[\text{Cl}^-]$ because an “off-diuretic-like” period never occurs.^{46,47}

If large amounts of alkali are ingestion or infused in a patient with acute or chronic kidney disease, this can produce metabolic alkalosis, which may sometimes be of extreme severity.⁴⁸ The “milk alkali” or “calcium alkali” syndrome is generated by the ingestion of large quantities of NaHCO_3 or other alkalinizing salts together with excess calcium (in milk or CaCO_3) intake.^{32,33} In these patients, kidney damage is partially generated by kidney calcium deposition, facilitated by high urine calcium concentrations in alkaline urine. Vomiting often also develops and will exacerbate the metabolic alkalosis. Citrate is an anticoagulant that is used in many blood products and is a HCO_3^- precursor. Each mole of $\text{Na}_3\text{Citrate}$ that is oxidized yields 3 moles of NaHCO_3 . Therefore, metabolic alkalosis can occur after massive blood transfusions and when blood products, such as fresh frozen plasma, are infused in large volumes during procedures such as plasma exchange. These metabolic alkaloses are usually maintained because GFR is reduced, which impairs kidney HCO_3^- excretion.

Compensation for chronic respiratory acidosis elevates plasma $[\text{HCO}_3^-]$ and reduces the $[\text{Cl}^-]$. However, arterial pH remains acid due to the elevated Pco_2 . If the chronically elevated Pco_2 is rapidly reduced, usually by mechanical ventilation, the plasma $[\text{HCO}_3^-]$ concentration may remain elevated for a period of time and generate metabolic alkalosis. This is especially likely if the patient has a low effective arterial blood volume or reduced GFR. This is “posthypercapnic” metabolic alkalosis. This alkalosis will persist until enough NaCl is ingested or infused to replete the extracellular fluid volume.⁴⁹

DIAGNOSTIC APPROACH TO METABOLIC ALKALOSIS

Most metabolic alkaloses are due to diuretics or external loss of gastric fluid. When the cause is not apparent from the history, drug review, and physical examination, it is helpful to categorize the disorder based on the kidney function and ECF or effective intraarterial volume status. If the GFR is reduced and major acidic gastrointestinal fluid losses do not exist, a source of exogenous bicarbonate should be sought. If the GFR is not markedly reduced, then it will be helpful to assess volume status integrating the history, physical examination, and spot urine $[\text{Cl}^-]$ measurement. Urine $[\text{Cl}^-]$ less than 20 mmol/L is consistent with a reduced ECF or effective intraarterial volume. Urine $[\text{Cl}^-]$ greater than 20 mmol/L suggests a volume expanded state, or defective kidney salt reabsorption (i.e., diuretic activity or kidney tubule reabsorption defect). Consider the diagnoses listed in Box 14.2. If the patient exhibits signs of volume expansion, such as hypertension, then consider the ECF volume-expanded disorders in Box 14.2. Measurements

of plasma renin and aldosterone levels can be helpful. If the patient appears volume contracted, yet has relatively high urine $[\text{Cl}^-]$, kidney salt wasting may be responsible. Many of the genetic mutations responsible for Bartter and Gitelman syndromes can be identified by commercial laboratories. Recognize that diuretic-generated metabolic alkalosis is generally characterized by cyclic changes in urine $[\text{Cl}^-]$ as the diuretic effects wax and wane, and widely varying urine $[\text{Cl}^-]$ generally indicates intermittent diuretic use (which some patients may deny). Fig. 14.6 shows this diagnostic approach.

TREATMENT

Treatment of the volume-contracted metabolic alkaloses, such as those due to the external loss of HCl from vomiting or NG suction, was discussed earlier. Intravenous NaCl is required to expand the contracted ECF and KCl to replete K^+ stores (the K^+ deficit is usually in the 200–400 mmol range). If the HCl losses continue, proton pump inhibitors can be used to reduce the acid loss. Treatment of metabolic alkalosis generated by loop and/or thiazide diuretics include stopping (or reducing the dose of) diuretics whenever feasible, administering KCl to replete K deficits, and adding blockers of distal tubule Na reabsorption (which also reduce H and K loss), such as triamterene or amiloride, or blocking mineralocorticoid activity with spironolactone or eplerenone. Adding or switching to acetazolamide, a diuretic that increases kidney bicarbonate excretion, can also rapidly correct the metabolic alkalosis but will increase kidney K loss. The combination of volume contraction, metabolic alkalosis, and hypokalemia, which characterizes Bartter and Gitelman syndromes, is difficult to manage. In addition to generous oral NaCl and KCl intake, drugs that blunt distal Na^+ reabsorption/ K^+ and H^+ exchange (e.g., spironolactone, eplerenone, or amiloride) are helpful, and nonsteroidal antiinflammatory drugs may be useful. The Gitelman phenotype usually also generates marked urine magnesium wasting and hypomagnesemia requiring aggressive magnesium replacement.

Management of metabolic alkalosis and hypokalemia due to excessive (and physiologically inappropriate) mineralocorticoid activity can be definitively treated by removing the source of mineralocorticoid (i.e., resect an adenoma when present). If this is not possible (i.e., bilateral adrenal hyperplasia), then spironolactone or eplerenone are very helpful. For disorders that mimic physiologically inappropriately excessive mineralocorticoid activity (i.e., Liddle syndrome), triamterene or amiloride are used (the excessive mineralocorticoid-like activity in these disorders is independent of aldosterone activity; therefore, spironolactone or eplerenone are not effective). All the physiologically inappropriate increased mineralocorticoid activity disorders, and the disorders that mimic this pathophysiology, are exacerbated by high NaCl intake and ameliorated with low NaCl intake. Administration of KCl to treat K depletion is also important.

Alkalosis driven by alkali ingestion is best treated by discontinuation of oral alkali and/or adjustment of dialysate bicarbonate in dialysis-dependent patients.

When severe metabolic alkalosis develops and cannot be corrected (e.g., dialysis or continuous kidney replacement cannot be initiated), then HCl can be infused. The HCl solution is prepared at a concentration of 100 to 150 mmol/L and must be infused into a central vein to avoid hemolysis and vein damage/sclerosis. The volume of distribution of the HCl is about 50% of body weight.

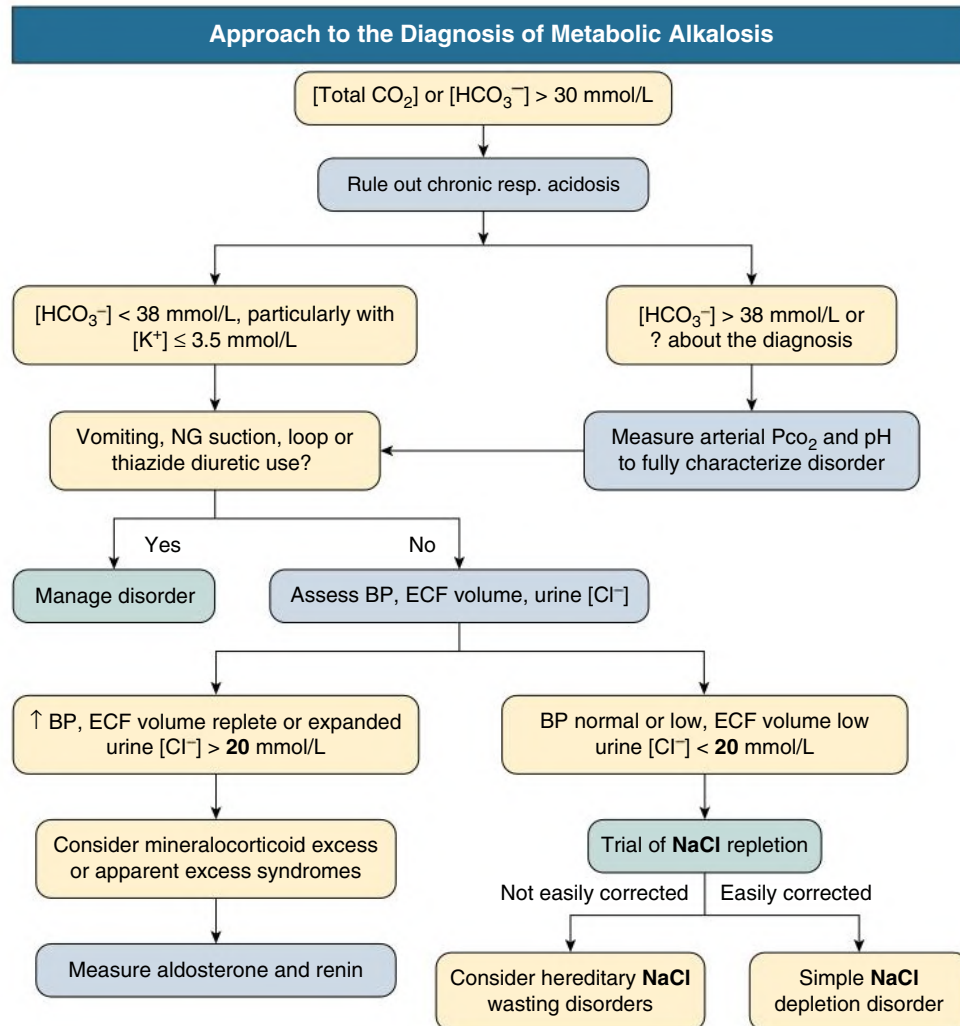


Fig. 14.6 Diagnostic approach to the causes of metabolic alkalosis. BP, Blood pressure; ECF, extracellular fluid; NG, nasogastric tube; Pco₂, partial pressure of carbon dioxide.

SELF-ASSESSMENT QUESTIONS

1. A 56-year-old man with a history of hypertension and chronic kidney disease stage 3b has required nasogastric suction for 2 days following right hemicolectomy for colonic carcinoma. His blood pressure (BP) is 110/70 mm Hg and pulse is 105 beats/min. There is no jugular venous distention. There are no bowel sounds noted and no leg edema. Laboratory values are Na 138, K 2.9, Cl 92, and total CO₂ 36 (all mmol/L). Creatinine is 2.4 mg/dL, and venous pH is 7.52. Which of the following laboratory studies would *most* help define the mechanism of his alkalosis?
 - A. Low urine chloride concentration
 - B. Low urine potassium concentration
 - C. Low urine sodium concentration
 - D. Urine pH greater than 6.0
2. In addition to recommending intravenous 0.9% saline for correction of metabolic alkalosis in the patient in question 1, he will also require which one of the following?
 - A. Addition of a proton pump inhibitor
 - B. HCl infusion
 - C. KCl infusion
 - D. Kidney replacement therapy
3. A 26-year-old woman presents to the emergency department reporting weakness. She has no significant past medical history. Examination reveals BP 112/76 mm Hg, pulse 96 beats/min, and 3/5 strength in both legs. Laboratory values are Na 142, K 2.6, Cl 96, and total CO₂ 34 (all mmol/L). Urine pH is 7, and urine electrolytes are Na 40, K 45, and Cl 10 (all mmol/L). The *most likely* cause of her illness is:
 - A. primary hyperaldosteronism.
 - B. Gitelman syndrome.
 - C. surreptitious vomiting.
 - D. diuretic use.

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